VERIFICATION AND CONJECTURES FOR LOG-CONCAVITY OF BRUHAT INTERVALS IN WEYL GROUPS

Abstract. Brenti conjectured that the rank sequence of every Bruhat interval in a finite Weyl group is log-concave. The restriction to Weyl groups is not incidental: the same statement is false for finite Coxeter groups in general, and Brenti gives a counterexample in the non-crystallographic group H_3. We verify the conjecture computationally for the Weyl groups of types A_2–A_7, B_2–B_6, D_4–D_6, E_6, F_4, and G_2. We checked 1,079,490,991 Bruhat intervals in exact integer arithmetic, and no interval violated log-concavity. This extends the previously recorded verification frontier by roughly an order of magnitude and completes the cases A_6, A_7, the full range of B_5, B_6, D_6, and E_6 left open in Brenti's list. We also report non-exhaustive evidence in higher rank: near-top slab sweeps in A_8–A_{10}, D_7–D_8, and E_7, together with 124,944 sampled and seeded intervals in B_7, B_8, D_7, and E_7.

These computations support, but do not prove, an extremal conjecture that in every irreducible, simply-laced Weyl group the global minimum of the log-concavity ratio over all Bruhat intervals is attained by the full interval $[e, w_0]$. If true, this would reduce verification of Brenti's conjecture for each such group from exponentially many Bruhat intervals to the single interval $[e, w_0]$, whose rank sequence is the coefficient sequence of the Poincaré polynomial. We show that both irreducibility and the simply-laced restriction are necessary. We prove the corresponding statement for rationally smooth lower intervals in the finite ranges stated in Theorem 5.5.

In type A, the ratio for $[e, w_0]$ is the central ratio of the Mahonian distribution. We prove an unconditional local limit theorem near the center for every m and determine the global minimum exactly, in integer arithmetic, for every $4 \leq m \leq 560$. The sharp asymptotic $\sigma_m^2(r_m - 1) = 1 - \frac{27}{25}m^{-1} + O(m^{-2})$ is conditional on a single explicitly stated core lemma for the exponentially tilted Mahonian distribution. This lemma is needed only for $m \geq 561$, and we reduce it, with explicit constants throughout, to six explicitly stated statements about the tilted law, one of which we have proved (the proof is recorded in the accompanying repository).

Finally, our computations find equality only in rank-two dihedral parabolic patterns of order $m \in \{4, 6\}$. We state this as an empirical classification and suggest, through the crystallographic restriction, why Weyl groups may avoid the failure in the non-crystallographic group H_3, whose counterexample interval sits on a dihedral core of order $m = 5$.

1. Introduction

Brenti's conjecture quantifies over every Bruhat interval of every finite Weyl group. This creates a computational obstacle and a structural question: direct verification requires an exponentially large family of interval checks, while the counterexample in H_3 shows that finiteness alone does not explain the Weyl-group case.

We checked 1,079,490,991 intervals in exact integer arithmetic. This is roughly an order of magnitude beyond the previously recorded frontier and completes the cases A_6, A_7, the full range of B_5, B_6, D_6, and E_6 left open in Brenti's list. We also formulate, for irreducible simply-laced Weyl groups, a conjectural reduction from all Bruhat intervals to the single interval $[e, w_0]$.

Let W be a finite Weyl group with simple reflections S, length function ℓ, and Bruhat order $\leq$. For $u \leq v$ in W, the *Bruhat interval* $[u, v] = \{z \in W : u \leq z \leq v\}$ carries a natural graded rank

Date: August 12, 2026.

2020 *Mathematics Subject Classification.* 05E15, 20F55, 06A07.

Key words and phrases. Bruhat order, Weyl groups, log-concavity, Mahonian numbers, Coxeter groups.

This investigation was carried out with substantial assistance from automated systems, and we omit a conventional byline rather than overstate the named researchers' personal contributions. *Nikol Panayotova Savova*, University of Oxford, nikol.p.savova@gmail.com; *Sihao Huang*, independent researcher, sihao.c.huang@gmail.com. They directed the work and take responsibility for it, and correspondence may be sent to either address.

sequence

$$a_k([u,v]) \;=\; \#\{z \in [u,v] : \ell(z) = \ell(u)+k\}, \qquad 0 \le k \le \ell(u,v) := \ell(v)-\ell(u),$$

and $[u,v]$ is *rank log-concave* if $a_k^2 \ge a_{k-1}a_{k+1}$ for every $1 \le k \le \ell(u,v)-1$. Brenti's Conjecture 2.11 in his 2024 survey of open problems on Coxeter groups and unimodality [1] asks whether this holds for *every* Bruhat interval of *every* finite Weyl group. The statement fails for general finite Coxeter groups, already in the non-crystallographic group H_3 (Section 2). Thus a proof for Weyl groups must use more than finiteness of the group.

Brenti records verification for types A_n and D_n with $n \le 5$, B_n with $n \le 4$, B_5 restricted to intervals of length at least 20, F_4, and the dihedral groups [1]. The top-heaviness inequality of Björner and Ekedahl [2] applies to parabolic lower intervals and constrains the *shape* of their rank sequences. It does not compare the ratios $a_k^2/(a_{k-1}a_{k+1})$ across intervals or identify where the log-concavity ratio is smallest.

The extremal conjecture is the structural proposal behind the paper. It predicts that, in each irreducible simply-laced Weyl group, the smallest ratio over all intervals already occurs at $[e,w_0]$. The statement remains conjectural for arbitrary intervals. If true, it replaces an exponentially large verification by a single check of the coefficient sequence of the Poincaré polynomial. We prove the comparison only for rationally smooth lower intervals in the finite ranges stated in Theorem 5.5.

In type A, that coefficient sequence is the Mahonian distribution. Under the proposed reduction, a Bruhat-order question becomes a question about one classical distribution, which is what makes the candidate extremal ratio tractable in type A. The type-A analysis also explains a trend in the data: the recorded minima in the type-A and type-D rows decrease toward 1 as rank grows. Taken alone, this can suggest that log-concavity will fail at a larger rank. Along the full type-A intervals, the central ratio instead approaches 1 from above at the rate determined in Section 6. We determine the global minimum over the Mahonian ratios exactly for every $4 \le m \le 560$; the sharp asymptotic for that global minimum beyond this finite range is conditional on Conjecture 6.3.

Contributions. Our contributions have four parts. We distinguish computational theorems, conditional results, and conjectural or empirical statements.

(i) Using exact integer arithmetic, we checked every Bruhat interval in types A_2–A_7, B_2–B_6 (the *full* range, not only $\ell(u,v) \ge 20$), D_4–D_6, E_6, F_4, and G_2. All 1,079,490,991 intervals were rank log-concave (Theorem 4.1). We also report non-exhaustive computations in higher rank: near-top slab sweeps reaching A_{10}, D_8, and E_7, and 124,944 sampled and targeted intervals in B_7, B_8, D_7, and E_7.

(ii) We conjecture that, in every irreducible, simply-laced Weyl group, the global minimum of the log-concavity ratio over *all* Bruhat intervals is attained by the full interval $[e,w_0]$ (Conjecture 5.1). Exhaustive low-rank computations and the near-top data support this conjecture but do not prove it in general. Theorem 5.5 proves the corresponding inequality for rationally smooth lower intervals in the finite ranges stated there. Two counterexamples show that irreducibility and the simply-laced restriction are both necessary.

(iii) In type A, we prove an unconditional local limit theorem for the Mahonian ratio near the center for every m, adapting the local central-limit-theorem technique of Canfield, Janson, and Zeilberger [3]. We determine the global minimum exactly, in integer arithmetic, for every $4 \le m \le 560$. Assuming one explicitly stated lemma about the exponentially tilted Mahonian distribution (Conjecture 6.3), we then prove the sharp asymptotic $\sigma_m^2(r_m-1) = 1 - \frac{27}{25}m^{-1} + O(m^{-2})$. The lemma is needed only for $m \ge 561$. We give a complete reduction with every constant explicit, and reduce the lemma in turn to six explicitly stated statements about the tilted law, one of which we have proved (Section 6; the proof and its interval certificate are recorded in the accompanying repository).

(iv) We give an empirical classification of the observed equality cases, in which log-concavity holds with equality, as rank-two dihedral parabolic patterns of order $m \in \{4, 6\}$. The crystallographic restriction theorem suggests a possible explanation for the contrast with H_3: its counterexample sits on a dihedral core of order $m = 5$, which is excluded from crystallographic reflection groups (Section 7).

Related work. In type A, the ratio for $[e, w_0]$ is a Mahonian ratio, linking the Bruhat-order question to a classical probability distribution. Log-concavity of the Mahonian numbers is classical: Bóna [4] gave a combinatorial proof, and it also follows from the product-closure results of [5, 6]. We do not claim that result here. For the related q-log-concavity of Gaussian binomial coefficients and of more general coefficient sequences, see [7, 8, 9]. Canfield, Janson, and Zeilberger [3] already obtain the central-ratio leading term $1 + \sigma^{-2}$ for Gaussian binomial coefficients. Our contribution is the S_m case: exact determination of the global minimum over the Mahonian ratios for every $4 \leq m \leq 560$, an unconditional local estimate for every $m \geq 4$, and, conditional on Conjecture 6.3, the sharp global asymptotic with explicit constants.

Several recent papers study related questions. Burrull, Gui, and Hu [10] prove asymptotic log-concavity for dominant lower intervals in *affine* Weyl groups under dilation. Their setting is affine, parabolic, and asymptotic in the dilation parameter, rather than the finite full-interval setting considered here. They also record a non-unimodal example due to Stanton showing that the direct parabolic analogue of Conjecture 2.11 can fail. Kessouri, Ahmia, Arslan, and Mesbahi [19] prove finite-n log-concavity of type-B q-Mahonian numbers but do not give an asymptotic statement, so their result does not overlap with our type-A asymptotic in (iii). Chapelier-Laget and Fromentin [20] recently refuted a different conjecture from the same survey. Our literature search, last updated on 2026-08-06, found no paper stating, proving, or refuting any of (i)–(iv).

Source code, exact-arithmetic result logs, and machine-checked numerical certificates for the proved claims are available at [repository URL to be added on submission]. These records make the finite verification reproducible. On the analytic side, Conjecture 6.3 isolates the single explicitly quantified input on which the sharp asymptotic depends, so future work can address that statement directly.

2. Preliminaries

Let W be a finite Coxeter group with simple generators S, length function ℓ, and Bruhat order $\leq$: $u \leq v$ iff some (equivalently, every) reduced word for v contains a reduced word for u as a subword. For a finite *Weyl* group W, i.e. the group generated by reflections in the root system of a Cartan matrix, we realize W as a subgroup of orthogonal transformations of a Euclidean space and compute every construction below directly from the Cartan matrix (Section 3).

For $u \leq v$ write $[u, v] = \{z \in W : u \leq z \leq v\}$ for the Bruhat interval, graded by $\ell(z) - \ell(u)$, with rank sequence

$$a_k([u, v]) = \#\{z \in [u, v] : \ell(z) = \ell(u) + k\}, \qquad 0 \leq k \leq \ell(u, v) := \ell(v) - \ell(u).$$

All $a_k([u, v]) > 0$ for Bruhat intervals, so we may define the *log-concavity ratio*

$$\rho([u, v]) = \min_{1 \leq k \leq \ell(u,v)-1} \frac{a_k([u, v])^2}{a_{k-1}([u, v])\, a_{k+1}([u, v])} \qquad (\ell(u, v) \geq 2),$$

and $[u, v]$ is rank log-concave exactly when $\rho([u, v]) \geq 1$. Brenti's Conjecture 2.11 [1] asserts $\rho([u, v]) \geq 1$ for every Bruhat interval of every finite *Weyl* group. Example 2.2 shows that the statement does not extend to every finite Coxeter group.

The interval $[e, w_0]$, where w_0 is the longest element, is distinguished: its rank sequence is exactly the coefficient sequence of the Poincaré polynomial $W(q) = \sum_{w \in W} q^{\ell(w)} = \prod_i [d_i]_q$ (product over the degrees d_i of W), so $a_k([e, w_0]) = [q^k]\, W(q)$.

Example 2.1 (Type A: Mahonian numbers)**.** For $W = S_m$ (type A_{m-1}), ℓ is the number of inversions, $W(q) = [m]_q! = \prod_{i=1}^{m}(1 + q + \cdots + q^{i-1})$, and $a_k([e, w_0]) = I_m(k) := \#\{\sigma \in S_m : \mathrm{inv}(\sigma) = k\}$, the classical Mahonian numbers. Log-concavity of $(I_m(k))_k$ itself is classical ([4]; also a consequence of [5, 6] applied to the factorization of $[m]_q!$); our contribution in Section 6 is the asymptotic behaviour of the *minimum ratio*, not log-concavity itself.

Example 2.2 (Failure for general Coxeter groups: H_3)**.** We checked the following example against the primary source. Let (W, S) be of type H_3, $S = \{s_1, s_2, s_3\}$ with $m(s_1, s_2) = 5$ and $m(s_2, s_3) = 3$, and let $u = s_3$, $v = s_1s_2s_3s_2s_1s_2s_1s_3$. Then [1, Conjecture 2.11] records the rank generating function of $[u, v]$ as

$$1 + 3t + 5t^2 + 7t^3 + 10t^4 + 10t^5 + 5t^6 + t^7.$$

Thus the rank sequence is $(a_0, \ldots, a_7) = (1, 3, 5, 7, 10, 10, 5, 1)$. It is unimodal but not log-concave: at $k = 3$, $a_3^2 = 49 < a_2a_4 = 50$, giving a margin of -1. The example appears in [1, Conjecture 2.11], *Some open problems on Coxeter groups and unimodality*, to appear in Proc. Sympos. Pure Math. **110**, on page 10 of the arXiv:2410.09897 PDF. The source gives the Coxeter parameters $m(s_1, s_2) = 5$, $m(s_2, s_3) = 3$, and $7^2 = 49 < 5 \cdot 10 = 50$. The rank-2 parabolic $\langle s_1, s_2\rangle$ generating the failure is dihedral of order $2m(s_1, s_2) = 10$, i.e. an $I_2(5)$ core. We use this fact in Section 7.

3. Computational methodology

The Bruhat-interval computations reported in Section 4, and the finite-range Mahonian computation of Theorem 6.2, use exact integer (arbitrary-precision) arithmetic throughout. The analytic certificates supporting the Section 6 reduction use directed-rounding interval arithmetic together with exact integer and rational arithmetic where stated; they are rigorous modulo the interval library (see the discussion following Proposition 6.8). We used three independent implementations and compared them with one another and with known invariants:

- The generic Cartan-matrix engine `weyl.py` builds any finite Weyl group directly from its Cartan matrix and computes $[u, v]$ by bitset-encoded up/down Bruhat-order BFS. We used it for the exhaustive tier.
- The complement-BFS engines `scaled.py` and `scaled_general.py` compute rank sequences of lower intervals $[e, v]$ with v close to w_0. They run BFS on the small complement $W \setminus [e, v]^{\downarrow}$ rather than on $[e, v]$ itself, allowing us to examine thin slabs in groups that cannot be enumerated exhaustively.
- We used the root-action engine `fast.py` for sampling and seeded perturbation searches.

We validated each engine against the following invariants: the known group order $|W|$; length $\equiv$ number of inversions (root-system definition) for every enumerated element; Poincaré polynomial $\equiv$ the degree-product formula $\prod_i [d_i]_q$; and, whenever two engines covered the same group, agreement on every interval.

The computations have three tiers:

Exhaustive: We enumerated and checked every Bruhat interval of the group. Only this tier gives a full verification of Conjecture 2.11 for the group in question (Theorem 4.1, Table 1).

Near-top slab: We checked only lower intervals $[e, v]$ with $\ell(w_0) - \ell(v)$ below a fixed bound. These computations provide evidence for Conjecture 2.11 and Conjecture 5.1 in the checked groups, but not a full verification.

Sampling / seeded: We checked random or targeted samples. These give coverage evidence only.

Remark 3.1 (E_6/B_6 segmentation and an unavailable witness)*.* We ran the E_6 and B_6 exhaustive computations in two complementary, non-overlapping segments, partitioned by the bottom element u, after an earlier single-pass process was killed. The documented segment boundaries cover the group, and the segment interval counts sum exactly to the totals in Table 1. An independent non-segmented full run also confirms the B_6 computation. For E_6, the first segment ($u < 6000$)

survived only as a running-minimum checkpoint stream, without the interval $[u, v]$ or rank sequence attaining that minimum. We therefore report the exact minimum ratio for E_6 but cannot exhibit a witness interval. A re-scan of $u < 6000$ remains a documented, low-cost follow-up; the missing witness does not affect Theorem 4.1.

The accompanying repository contains the raw result logs, run records, and CI configuration for reproducing the exhaustive tier ([repository URL to be added on submission]).

4. Verification results

Table 1. Exhaustive tier: every Bruhat interval, exact integer arithmetic.

Group	$\|W\|$	#intervals ($\ell \geq 2$)	min ratio	witness (index k)
A_2	6	19	2.000000	$[e, 121]$, $k = 1$
A_3	24	213	1.388889	$[e, 12321]$, $k = 2$
A_4	120	3,781	1.210000	$[e, w_0]$, $k = 5$
A_5	720	98,407	1.122222	proper (ties $[e, w_0]$), $k = 7$
A_6	5,040	3,550,919	1.079096	proper (ties $[e, w_0]$), $k = 10$
A_7	40,320	170,288,585	$1.054250 = 919681/872356$	proper, $\ell = 27$ (ties $[e, w_0]$), $k = 14$
B_2	8	33	1.000000	$[e, 1212]$, $k = 2$
B_3	48	847	1.000000	$[e, 2323]$, $k = 2$
B_4	384	40,249	1.000000	$[e, 3434]$, $k = 2$
B_5	3,840	3,089,459	1.000000	$[e, 4545]$, $k = 2$
B_6	46,080	350,676,009	1.000000	$[e, 5656]$, $k = 2$
D_4	192	9,817	1.136232	proper (ties $[e, w_0]$), $\ell = 11$, $k = 5$
D_5	1,920	745,377	1.069459	proper (ties $[e, w_0]$), $\ell = 19$, $k = 10$
D_6	23,040	84,339,681	1.040703	proper, $\ell = 27$ (ties $[e, w_0]$), $k = 15$
E_6	51,840	466,250,713	1.028446	witness unrecoverable, see Remark 3.1
F_4	1,152	396,809	1.000000	$[e, 2323]$, $k = 2$
G_2	12	73	1.000000	$[e, 1212]$, $k = 2$

Theorem 4.1 (Exhaustive tier)**.** *Let W be one of the Weyl groups $A_2, \dots, A_7$, $B_2, \dots, B_6$, D_4, D_5, D_6, E_6, F_4, G_2. Then every Bruhat interval $[u, v] \subseteq W$ is rank log-concave. The verification is exhaustive: we enumerated and checked every interval of length ≥ 2 in exact integer arithmetic (Table 1). In total, we checked* 1,079,490,991 *intervals, none of which violated log-concavity.*

Remark 4.2. Compared with Brenti's recorded list (A_n, D_n for $n \leq 5$; B_n for $n \leq 4$; B_5 only for $\ell(u, v) \geq 20$; F_4; dihedral groups), our exhaustive computation adds A_6, A_7, B_5 (full range), B_6, D_6, and E_6.

Theorem 4.3 (Near-top tier — lower intervals only; not exhaustive)**.** *For each pair (W, c) below, every lower interval $[e, v]$ with $\ell(w_0) - \ell(v) \leq c$ is rank log-concave; moreover the minimum of $\rho([e, v])$ over this slab is attained at $v = w_0$ (Table 2):*

$$(A_7, 4),\ (A_8, 4),\ (A_9, 3),\ (D_7, 3),\ (D_8, 3),\ (E_7, 2).$$

In A_{10}, $[e, w_0]$ and a partial slab (11 of 65 candidates at $c = 2$) were checked, all log-concave, with one proper interval exactly tying $\rho([e, w_0])$.

Remark 4.4. The two exact rational minima in Table 2 are $\rho([e, w_0]) = 919681/872356$ in A_7 and $\rho([e, w_0]) = 65523/64757$ in E_7. The E_7 rank sequence sums to 2,903,040 $= |W(E_7)|$, confirming that it is the full interval rather than a shorter proper interval.

TABLE 2. Near-top slab tier: lower intervals $[e, v]$, $\ell(w_0) - \ell(v) \le c$ (not exhaustive). Exact fractions for the min ratio, where computed, are given in the text.

Group	$\|W\|$	slab	#checked	min ratio	at k
A_7	40,320	$c \le 4$	full	1.054250	$[e, w_0]$, 14
A_8	362,880	$c \le 4$	full	1.038942	$[e, w_0]$, 18
A_9	3,628,800	$c \le 3$	209/209	1.028950	$[e, w_0]$, 22
A_{10}	39,916,800	$c \le 2$	11/65*	1.022102	tie†, 27
D_7	322,560	$c \le 3$	112/112	1.025574	$[e, w_0]$, 21
D_8	5,160,960	$c \le 3$	156/156	1.017122	$[e, w_0]$, 28
E_7	2,903,040	$c \le 2$	full	1.011829	$[e, w_0]$, 31

*partial, resumable; †$[e, w_0]$ attains the minimum and one proper interval exactly ties it.

Remark 4.5. Theorem 4.3 concerns only lower intervals in a thin slab below w_0. It provides evidence for Conjecture 2.11 and Conjecture 5.1, including in E_7, but verifies neither conjecture beyond the stated slab.

TABLE 3. Sampling / seeded tier (not exhaustive; coverage counts).

Probe	Group	#intervals	design	min ratio	note
random (seed 7)	B_7	20,000	$4 \le \ell(u, v) \le 12$	1.000000	$(1, 2, 2, 2, 1)$ pattern
random (seed 7)	D_7	20,000	same	$1.157143 = 81/70$	proper, k=4
random (seed 7)	E_7	20,000	same	$1.142400 = 14641/12816$	proper, k=5
seeded (seed 3)	B_7	1,324	m=4 core + 0–6 covers	1.000000 only at pure core	28 ratio-1 hits
seeded (seed 4)	B_7	32,458	m=4 core + 0–10 covers	pert.≥ 1: ≥ 1.079153	margins 4–8%
seeded (seed 4)	B_8	31,162	m=4 core + 0–10 covers	pert.≥ 1: ≥ 1.087990	margins 4–8%

Remark 4.6. For the seeded probes, we attempted a fixed number of perturbations of the $m = 4$ dihedral core: 4,000, 100,000, and 100,000 in the three runs above. We discarded an attempt whenever the targeted perturbation did not realize a Bruhat up-cover chain. The #intervals column counts only the intervals that we actually checked.

Proposition 4.7 (Sampling tier — not exhaustive; coverage statement only)**.** (i) *We sampled* 60,000 *Bruhat intervals (*20,000 *each in B_7, D_7, and E_7) uniformly from a random-walk ensemble with* $4 \le \ell(u, v) \le 12$ *(seed* 7*, reproducible). All are rank log-concave.* (ii) *We checked* 64,944 *seeded intervals containing an $m = 4$ dihedral braid core perturbed by* 0–10 *extra cover steps. All are rank log-concave, and every perturbation that destroys the pure dihedral pattern strictly raises the ratio (Table 3). The seeded search attempted more perturbations, but we discarded those that did not realize a genuine cover chain; see the remark following the table.*

In total, we checked 124,944 intervals in the sampling and seeded probes. All were rank log-concave, and equality occurred only in the $(1, 2, \ldots, 2, 1)$ pattern discussed in Section 7.

5. AN EXTREMAL CONJECTURE

Conjecture 5.1 (F1)**.** *Let W be a finite Weyl group that is irreducible and simply-laced (type A_n, D_n, E_6, E_7, or E_8), with longest element w_0. Then*

$$\min_{\substack{[u,v]\subseteq W\\ \ell(u,v)\ge 2}} \rho([u, v]) \;=\; \rho([e, w_0]) \;=\; \min_{1\le k\le \ell(w_0)-1} \frac{b_k^2}{b_{k-1}b_{k+1}},$$

where $b_k = [q^k]\,W(q)$*: the global minimum of the log-concavity ratio over all Bruhat intervals of* W *is attained by the full interval* $[e, w_0]$*, at a central index. Proper intervals may tie this minimum (observed for* A_5, A_6, A_7, D_6 *in the exhaustive tier, and for a further proper interval in the* A_{10} *near-top slab) but never fall below it.*

The following two counterexamples show that both hypotheses in Conjecture 5.1 are necessary.

Example 5.2 (Simply-laced is necessary)**.** In B_3, the interval $[e, (s_2s_3)^2]$ has rank sequence $(1, 2, 2, 2, 1)$ and $\rho = 2^2/(2 \cdot 2) = 1$ exactly, on a plateau, strictly below $\rho([e, w_0(B_3)]) = 8/7$. The analogous B_4 interval $[e, (s_3s_4)^2]$ gives $\rho = 1 < 968/897 = \rho([e, w_0(B_4)])$. The interval $[e, (s_2s_3)^2]$ is *rationally smooth* (its rank sequence is palindromic), so dropping the simply-laced hypothesis falsifies the conjecture even within the rationally-smooth subclass discussed below.

Example 5.3 (Irreducible is necessary)**.** Let $W = A_1 \times D_4$ (simply-laced, reducible) and $v = (e, w_0(D_4)) \in W$. The interval $[e, v]$ is isomorphic, as a graded poset, to $[e, w_0(D_4)]$ inside the D_4 factor, with rank sequence $P_{D_4} = (1, 4, 9, 16, 23, 28, 30, 28, 23, 16, 9, 4, 1)$ and $\rho([e, v]) = 28^2/(23 \cdot 30) = 392/345 = 1.1362318\ldots$ at $k = 5$. This is a *proper* interval of W: $\ell(v) = 12 < 13 = \ell(w_0(W))$. The full interval has rank sequence $[2]_q \cdot P_{D_4} = (1, 5, 13, 25, 39, 51, 58, 58, 51, 39, 25, 13, 5, 1)$ and $\rho([e, w_0(W)]) = 58^2/(51 \cdot 58) = 58/51 = 1.1372549\ldots$ Since $392 \cdot 51 = 19992 < 20010 = 345 \cdot 58$,

$$\rho([e, v]) \;=\; \frac{392}{345} \;<\; \frac{58}{51} \;=\; \rho([e, w_0(W)]), \qquad \text{gap } \frac{58}{51} - \frac{392}{345} = \frac{2}{1955} \approx 0.00102.$$

Thus a *proper* Bruhat interval of the reducible simply-laced group $A_1 \times D_4$ has a smaller ratio than $[e, w_0]$. The extremal principle therefore fails when reducibility is allowed. The witness v is smooth, but the example already refutes the unrestricted statement for reducible W, independently of smoothness. In an exhaustive check of all 27 reducible simply-laced types of rank ≤ 6, $A_1 \times D_4$ was the *only* group for which the inequality failed.

Remark 5.4 (Status: conjectural, not proved)*.* Conjecture 5.1 is not proved for arbitrary $[u, v]$ in any infinite family. We verified it exhaustively for every irreducible simply-laced group in the exhaustive tier (A_2–A_7, D_4–D_6, E_6; Table 1). In each case $[e, w_0]$ attains the minimum; in A_5, A_6, A_7, D_6, a specific proper interval ties it. We also verified the conjectured comparison in the near-top slabs for A_8–A_{10}, D_7–D_8, and E_7 (Table 2). These finite computations do not prove the conjecture in general rank.

Theorem 5.5 (F1 restricted to rationally smooth intervals)**.** *Let* W *be a finite irreducible simply-laced Weyl group of rank* ≤ 6 *(*$A_1, \ldots, A_6, D_4, D_5, D_6, E_6$*). For every rationally smooth* $v \in W$*, i.e. every* v *for which the Poincaré polynomial* $P_{[e,v]}(q)$ *factors as a product of q-integers* [11, 12]*, we have* $\rho([e, v]) \ge \rho([e, w_0])$*, with equality iff* $v = w_0$*. Granting the classical type-A factorization result of Gasharov* [12]*, the same conclusion holds in type* A_{m-1} *for* $m \le 17$*. Our working notes verify that factorization computationally only through* $m \le 7$*; for the remaining cases we rely on the cited result rather than re-derive it.*

Conjecture 5.6 (Staircase domination)**.** *In type* A_{m-1}*, the exponent multiset of the rationally-smooth Poincaré polynomial* P_v *is always a gap-free ("staircase") multiset dominated (coefficient-wise, after sorting) by the degree multiset* $\{2, \ldots, m\}$*.*

Remark 5.7*.* We verified Conjecture 5.6 on 91,355 dominated pairs and, for the instances used in Theorem 5.5, on a further 131,036 pairs with $m \le 17$. It remains unproved for general m. If it holds, the $m \le 17$ restriction in the type-A clause of Theorem 5.5 can be removed, giving the same inequality for every m.

Remark 5.8*.* Theorem 5.5 proves a *strictly weaker* statement than Conjecture 5.1: it compares $\rho([e, v])$ with $\rho([e, w_0])$ only when v is rationally smooth, not for arbitrary v. Examples 5.2 and 5.3 show that the irreducible and simply-laced hypotheses remain necessary for this restricted statement; the B_3 counterexample is itself rationally smooth. Type D_n for $n \geq 7$ and types E_7, E_8 remain open even for this subclass. Extending this theorem is one possible route toward Conjecture 5.1.

6. The type-A case: Mahonian asymptotics

Throughout this section $W = S_m$ (type A_{m-1}), $I_m(k) = \#\{\sigma \in S_m : \mathrm{inv}(\sigma) = k\}$ so that $a_k([e, w_0]) = I_m(k)$, $N = \binom{m}{2}$, $\sigma^2 = \mathrm{Var}(\mathrm{inv}) = m(m-1)(2m+5)/72$, $r_m(k) = I_m(k)^2/(I_m(k-1)I_m(k+1))$, and $r_m = \min_{1 \leq k \leq N-1} r_m(k)$. Write $y = (k - N/2)/\sigma$, and let $B_m := (S_4 - m)/(240\sigma^4)$ where $S_r = \sum_{j=1}^m j^r$, so $B_m = \frac{27}{25} m^{-1}(1 + O(m^{-1}))$.

What is proved unconditionally. We prove the following local limit theorem for the log-concavity ratio, in the classical style of [14], near the center for *every* m. Every constant is explicit. The proof combines an analytic argument with machine-verified finite checks, and two independent automated review passes (an adversarial mathematics lane and a numerics lane) examined the mathematics and numerics separately.

Theorem 6.1 (Pointwise and window law for the Mahonian ratio)**.** *Let $p(k) = I_m(k)/m!$, $Z(y) = (2\pi\sigma^2)^{-1/2} e^{-y^2/2}$, and let He_4 denote the fourth probabilists' Hermite polynomial.*

(a) (Pointwise.) *For every $m \geq 4$ and every $0 \leq k \leq N$,*

$$\left| p(k) - Z(y)\left[1 - \tfrac{B_m}{12} \mathrm{He}_4(y)\right] \right| \;\leq\; \frac{0.45}{\sigma m^2}.$$

(b) (Window ratio law.) *For every fixed $y_0 \geq 0$ there are explicit $C_2(y_0)$ and $m_1(y_0)$ (e.g. $C_2(1) = 3.1$, $m_1(1) = 180$; the constant grows with y_0, e.g. $C_2(3) = 3940$) such that for every $m \geq m_1(y_0)$ and every k with $|y| \leq y_0$,*

$$\sigma^2 \log r_m(k) \;=\; 1 + B_m(y^2 - 1) + E_1(k), \qquad |E_1(k)| \leq \frac{C_2(y_0)}{m^2}.$$

We also verified (a), with the same displayed constant, directly in directed-rounding interval arithmetic over the exact integer Mahonian coefficients, for every $4 \leq m \leq 109$. We verified the window law in (b), with the same displayed constants, for every $4 \leq m \leq 229$. The analytic argument covers all remaining values of m. Thus these finite ranges are independent checks rather than thresholds, and the window law in (b) is unconditional throughout every range used below.

Theorem 6.1 determines the central ratio exactly to the stated order: at $y = 0$, part (b) gives

$$\sigma^2\big(r_m(\lfloor N/2 \rfloor) - 1\big) \;=\; 1 - \tfrac{27}{25} m^{-1} + O(m^{-2}). \tag{1}$$

Exact Mahonian data at $m = 50$ agree with this expansion to four significant digits.

We also determine the location and value of the global minimum, together with the sharp finite-m bound, exactly for every $4 \leq m \leq 560$.

Theorem 6.2 (Exact finite range)**.** *For every $5 \leq m \leq 560$, with every verdict computed in exact integer or rational arithmetic:*

(i) the minimum of $r_m(k)$ over $1 \leq k \leq N-1$ is attained at the central index $k = \lfloor N/2 \rfloor$ (for N odd, the mirror index $\lceil N/2 \rceil$ ties it exactly, as palindromy forces);
(ii) consequently $r_m = r_m(\lfloor N/2 \rfloor)$;
(iii) $\sigma^2(r_m - 1) \geq \frac{187}{216}$, with equality if and only if $m = 6$;
(iv) $m \mapsto \sigma^2(r_m - 1)$ is strictly increasing for $6 \leq m \leq 560$, reaching $\sigma^2(r_m - 1) = 0.998072\ldots$ at $m = 560$, where $m\big(1 - \sigma^2(r_m - 1)\big) \approx 1.07935$, consistent with the limit $\frac{27}{25} = 1.08$.

At $m = 4$, and only there in the computed range, the minimizing index is not central ($\mathrm{argmin} = 2$, *$N/2 = 3$) and $\sigma^2(r_4 - 1) = \frac{91}{108} < \frac{187}{216}$.*

We computed the coefficients with a checkpointed implementation of the q-factorial recurrence for I_m. Every comparison uses integer or exact rational arithmetic; floating point appears only in printed displays. The repository contains the complete result logs, with one row for every $4 \le m \le 560$, all passing.

From the center to the global minimum: the reduction. Equation (1) controls the ratio at and near the center, and Theorem 6.2 settles every $4 \le m \le 560$. For $m \ge 561$, the sharp asymptotic also requires ruling out a smaller ratio elsewhere in the bulk. We prove the required comparison conditional on one estimate for the exponentially tilted Mahonian distribution in the deep-tilt regime, stated as Conjecture 6.3. Every constant in the reduction is explicit.

For $\lambda \in \mathbb{R}$ let $U_1^\lambda, \dots, U_m^\lambda$ be independent, with U_j^λ supported on $\{0, 1, \dots, j-1\}$ and $\Pr(U_j^\lambda = i) \propto e^{-\lambda i}$, and set $S_\lambda := \sum_{j=1}^m U_j^\lambda$ (not to be confused with the power sums S_r above), $\mu(\lambda) := \mathbb{E} S_\lambda$. At $\lambda = 0$ this is the inversion statistic: $\Pr(S_0 = k) = I_m(k)/m!$. Since $\mu'(\lambda) = -\operatorname{Var} S_\lambda < 0$, every interior k ($1 \le k \le N-1$) has a unique *mean-matching tilt* $\lambda(k)$ with $\mu(\lambda(k)) = k$; write $s^2 = s^2(k) := \operatorname{Var} S_{\lambda(k)}$, let κ_3, κ_4 denote the third and fourth cumulants of $S_{\lambda(k)}$, and let $\varphi(t) := \mathbb{E}\, e^{it(S_{\lambda(k)} - k)}$. By palindromy $r_m(N-k) = r_m(k)$ and $\lambda(N-k) = -\lambda(k)$, so we take $k \le N/2$, i.e. $\lambda(k) \ge 0$, throughout, and set $\omega := m\lambda(k)$.

Conjecture 6.3 (Core lemma $\mathrm{CL}(79, 20, 0.89)$)**.** *For every $m \ge 561$ and every interior k with $s^2(k) \ge 79$ and $4/m < \lambda(k) \le 0.89$,*

$$r_m(k) - 1 \;=\; \frac{1}{s^2}\Big(1 + \theta\, \frac{20}{\min(m, s^2)}\Big) \qquad \textit{for some } \theta = \theta(m,k) \in [-1, 1].$$

Remark 6.4. Three points clarify the statement. (a) The hypothesis $s^2 \ge 79$ is automatic on the stated tilt range, where $s^2 \ge 1122800/7921 > 141$. (b) Theorem 6.5 uses only the lower-bound half, corresponding to $\theta \ge -1$. (c) The conjecture excludes the complementary small-tilt range $\lambda(k) \le 4/m$ because the proved window law covers it unconditionally; see region (d) of Remark 6.6. The reduction for Theorem 6.5 requires this statement for every $m \ge 401$. The exact computation in Theorem 6.2 handles $401 \le m \le 560$, so the conjecture is stated and needed only for $m \ge 561$.

Theorem 6.5 (F2, sharp asymptotic — conditional)**.** *Assume Conjecture 6.3. Then*

$$\sigma^2(r_m - 1) \;=\; 1 - \tfrac{27}{25} m^{-1} + O(m^{-2}), \qquad \textit{equivalently} \qquad r_m - 1 \sim \frac{36}{m^3},$$

and in particular $\sigma^2(r_m - 1) \to 1$. The error term is explicit and two-sided: for every $m \ge 561$,

$$1 - B_m - \frac{C_A}{m^2} \;\le\; \sigma^2(r_m - 1) \;\le\; 1 - B_m + \frac{1.8}{m^2}, \qquad C_A = 37997.85,$$

the upper bound holding unconditionally for every $m \ge 180$.

Remark 6.6 (Shape of the reduction, and where the conjecture enters). We summarize the reduction; the repository contains the explicit constants and machine-verified certificates for each step (directed-rounding interval arithmetic together with exact rational arithmetic where stated). Let $m \ge 561$. By palindromy, take $k \le N/2$. We divide the interior indices into four regions according to k and $\omega(k) = m\lambda(k)$, with $K_c := \min(\frac{7}{10}m,\, m-1)$ for $m < 1581$ and $K_c := m - 1$ beyond:

(a) $k = 1$: an exact pentagonal-recurrence bound gives $r_m(1) - 1 \ge \frac{m-1}{2(m+1)}$, hence $\sigma^2(r_m(1) - 1) \ge 10^5$;
(b) $2 \le k \le K_c$: an elementary two-term bound gives $r_m(k) - 1 \ge \frac{m-1}{2k(m+k)}$, hence $\sigma^2(r_m(k) - 1) \ge 1879$;
(c) $k > K_c$ with $\omega(k) > 4$ (the deep-tilt region): here $\min(m, s^2) \ge 79.5$, $s^2 \le 0.72711\, \sigma^2$, and $\lambda(k) \le \log(17/7) < 0.89$, all proved. This is the only point at which we use Conjecture 6.3; it gives $\sigma^2(r_m(k) - 1) \ge \big(1 - \frac{20}{79.5}\big)/0.72711 \ge 1.0293$;

(d) $k > K_c$ with $\omega(k) \le 4$: the window law of Theorem 6.1, extended unconditionally and with explicit constants to the full small-tilt window $|\omega| \le 4$, gives $\sigma^2(r_m(k) - 1) \ge 1 - B_m - C_A/m^2$.

The bounds in regions (a)–(c) all exceed $1.02 > 1 - B_m$, so the global minimum lies in region (d). This gives the lower bound. The upper bound follows from $r_m \le r_m(\lfloor N/2 \rfloor)$ and the explicit form of (1). The value of C_A comes from one triangle-inequality estimate; numerical evaluation places the corresponding dominant contribution near 5. Any fixed constant suffices for the sharp asymptotic.

Reduction of the core lemma to six statements. We reduce Conjecture 6.3, with explicit constants throughout, to six statements about the tilted law, five of which remain open. Partition the deep-tilt values $\omega = m\lambda \in (4,\, 0.89\,m]$ into seven bands

$$\mathcal{W}_1 = (4,5],\ \mathcal{W}_2 = (5,6],\ \mathcal{W}_3 = (6,8],\ \mathcal{W}_4 = (8,10],\ \mathcal{W}_5 = (10,20],\ \mathcal{W}_6 = (20,40],\ \mathcal{W}_7 = (40,\infty),$$

each intersected with $\{\lambda \le 0.89\}$, and normalize the cumulants on the tilt scale:

$$\tilde\kappa_3 := \frac{|\kappa_3|\,\lambda}{s^2}, \qquad \tilde\kappa_4 := \frac{\kappa_4\,\lambda^2}{s^2}.$$

Table 4. Per-band constants in Conjecture 6.7 (bands of $\omega = m\lambda$). The J_0 entries are six-digit displays of exact rational constants archived in the repository.

band ω-range	$\mathcal{W}_1$ $(4,5]$	$\mathcal{W}_2$ $(5,6]$	$\mathcal{W}_3$ $(6,8]$	$\mathcal{W}_4$ $(8,10]$	$\mathcal{W}_5$ $(10,20]$	$\mathcal{W}_6$ $(20,40]$	$\mathcal{W}_7$ $(40,\infty)$
R_3	1.0	1.2	1.5	1.7	2.0	2.1	2.2
R_4	0.8	1.4	2.6	3.5	5.2	6.0	6.6
C_5	0.05	0.06	0.08	0.10	0.15	0.25	0.80
J_0	0.682942	1.10268	1.91562	2.53645	3.66793	4.17806	4.59597

Conjecture 6.7 (The supporting statements; (S1) is now a theorem)**.** *For every $m \ge 561$ and every interior k with $\lambda = \lambda(k) \in (4/m,\, 0.89]$, with $\mathcal{W}$ the band containing $\omega = m\lambda$ and constants R_3, R_4, C_5, J_0 as in Table 4:*

- *(S1) (banded cumulant scales; **proved**, see Remark 6.9) $\tilde\kappa_3 \le R_3(\mathcal{W})$ and $\tilde\kappa_4 \le R_4(\mathcal{W})$;*
- *(S2) (fifth-order core remainder) for $0 \le t \le \lambda/2$, with the continuous branch of $\log\varphi$ starting at $\log\varphi(0) = 0$,*
$$\left|\log\varphi(t) + \frac{s^2t^2}{2} + \frac{i\kappa_3 t^3}{6} - \frac{\kappa_4 t^4}{24}\right| \;\le\; C_5(\mathcal{W})\,\frac{s^2t^5}{\lambda^3};$$
- *(S3) (joint cancellation) $\tilde\kappa_3^2 - \tilde\kappa_4/2 \le J_0(\mathcal{W})$;*
- *(S4) (a-priori ratio seed) $\left|s^2\big(r_m(k)-1\big) - 1\right| \le 0.89$;*
- *(S5) (ω-continuum certificate) the band-$\mathcal{W}_1$ row bound holds for every real $\omega \in (4,5]$ and every integer $561 \le m \le 699$, not merely at sampled ω;*
- *(S6) (bootstrap closure) the self-consistency map used in the proof of Proposition 6.8 satisfies $G(t) < t$ throughout $20/m < t \le 0.89$, uniformly in m, in the band, and in λ.*

Proposition 6.8 (Reduction of the core lemma)**.** *Statements (S1)–(S6) of Conjecture 6.7 imply Conjecture 6.3 — that is, the composed chain delivers exactly the constant* 20 *of* $\mathrm{CL}(79, 20, 0.89)$ *— for every $m \ge 561$.*

The proof of Proposition 6.8 proceeds band by band through Table 4. Statements (S1) and (S3) bound the main Gaussian-model term, (S2) bounds the core remainder, and (S4) starts a self-consistency argument that closes, by (S6), through strict contraction. Statement (S5) supplies the continuum certificate for the lowest band. All other ingredients, including the variance floors and caps, the far-region decay of φ, and the crossover and mid-range bounds, are proved with explicit constants. The reduction argument is not reproduced here. The repository contains the band constants, the per-band certificates, and the composition document; the bootstrap map of (S6) is not yet specified in closed form there, which is part of why (S6) remains open, and the assembled composition has not yet been independently refereed. Those computations use directed-rounding interval arithmetic together with exact integer and rational arithmetic where stated; they are rigorous modulo the interval library, and we do not claim they are exact-rational throughout.

Remark 6.9 (Status and evidence for (S1)–(S6)). Statement (S1) has since been proved, by a bandwise argument with a rigorous interval certificate; the argument and its certificate are recorded in the accompanying repository. We record it here as part of the reduction because the remaining statements are stated relative to the same bands and constants. The other five remain open, and the proof of Proposition 6.8 uses each of them. The list has changed as the reduction was scrutinised: two earlier hypotheses were proved and removed; (S3) and (S4) were added after simpler arguments failed; and (S5) and (S6) were added after an adversarial review found that the composition step had been relying on a finite sample of ω and on a fixed-point ansatz respectively. We regard the current list as the honest one, while noting that it has grown under scrutiny before. We report the numerical evidence and its margins below.

- (S1): At the deep corner of $\mathcal{W}_7$, the computed suprema of $\tilde{\kappa}_3$ and $\tilde{\kappa}_4$ are 2.1215 and 6.3552 (large-m limits 2.1303 and 6.4113), compared with the bounds 2.2 and 6.6. The margins are only 3.7% and 3.9%, the smallest in the reduction.
- (S2): The computed remainder ratios lie between 0.0083 and 0.2104. Depending on the band, the bounds C_5 exceed these values by a factor between 1.6 and 8. For the lowest band, this comparison was made at interior probe points rather than uniformly up to the edge.
- (S3): The smallest computed margin is 32.6%, at $(m, \omega) = (561, 5.0)$. A point consistent with (S1) and $\kappa_4 \geq 0$ violates the estimate supplied by (S3) by 43%. Therefore the present argument requires a joint bound of this form; a separate treatment of the two cumulants does not suffice.
- (S4): This is a weak a-priori form of the conclusion, with constant 0.89 where Conjecture 6.3 gives $20/\min(m, s^2) \leq 0.036$, since $s^2 \geq m$ on this range. We state it separately because it has not been proved. Machine checks, in directed-rounding interval arithmetic, of the inequality in Conjecture 6.3 at $m = 401$ and $m = 402$, below its stated range, cover 260 adversarially chosen interior indices and have a $17\times$ margin. Every tilted ratio computed in this project also lies well within 0.89.
- (S5): The per-cell floors give a computed row bound of 0.4165 at $m = 561$ and 0.2601 at $m = 699$, against the required threshold 1. What is missing is not margin but coverage: the available checks are a finite ω-grid, and the monotonicity available to us runs in a different variable, so it does not interpolate between sampled points.
- (S6): The map satisfies $G(20/m) < 20/m$ at the two thinnest rows checked ($0.0492 < 0.0499$ and $0.0421 < 0.0432$); both rows lie below the range in which (S6) is needed, so a domination argument to the stated range is among what is missing. The measured basin extends to about 0.90. An explicit closed form for the map G, convexity of the branches, directed bounds at both endpoints, and a reduction showing that the thinnest rows dominate every band and every $m \geq 561$ are all still required.

Remark 6.10 (What is claimed, and what is not). Theorems 6.1 and 6.2 are unconditional, as is the upper-bound half of Theorem 6.5. The sharp asymptotic in Theorem 6.5 assumes only Conjecture 6.3. Proposition 6.8 proves that (S1)–(S6) imply Conjecture 6.3. Statement (S1) is proved;

(S2)–(S6) remain open. In particular, the numerical margins in Remark 6.9 do not constitute proofs of the open statements; for (S5) the gap is coverage rather than margin.

Remark 6.11 (Location and explicit constant). Two further statements concern the minimizing index and the finite-m bound. The minimizing index is exactly central for every $5 \le m \le 560$ (Theorem 6.2). For $m \ge 561$, conditional on Conjecture 6.3, the reduction places it in the small-tilt window $|\lambda(k)| \le 4/m$ around the center; see region (d) of Remark 6.6. The stronger statement $|\mathrm{argmin} - N/2| \le 1$ for every m remains open. It is equivalent to a third-difference positivity property of $(I_m(k))_k$ and has been verified for $5 \le m \le 56$, but we do not have a proof.

The explicit bound $r_m \ge 1 + \frac{187}{216}\sigma_m^{-2}$ for all $m \ge 5$ is sharp at $m = 6$; the earlier guess $7/8$ already fails at $m = 6$. The bound is exact for $5 \le m \le 560$ by Theorem 6.2(iii). It holds for every $m \ge 561$ conditional on Conjecture 6.3, because the explicit lower bound in Theorem 6.5 exceeds $\frac{187}{216}$ for every $m \ge 537$. Thus, assuming the conjecture, the sharp bound holds for every $m \ge 5$, with equality only at $m = 6$.

7. Equality cases

Observation 7.1 (F3, equality classification)**.** Consider the Bruhat intervals $[u, v] \subseteq W$ in our exhaustive and seeded-perturbation data, where W is a finite Weyl group, $\ell(u, v) \ge 2$, and $a_k^2 = a_{k-1}a_{k+1}$ for some interior k. In every observed case, the full rank sequence of $[u, v]$ is

$$(a_0, \dots, a_{\ell(u,v)}) \;=\; (1, 2, 2, \dots, 2, 1) \qquad (m-1 \text{ interior 2's},\; m \ge 4).$$

Each observed interval is therefore poset-isomorphic to the full Bruhat interval of the dihedral group $I_2(m)$ and arises from a rank-two dihedral standard parabolic of braid order m. The crystallographic restriction confines rank-two parabolics of Weyl groups to $m \in \{2, 3, 4, 6\}$. Since equality requires $m \ge 4$, the observed equality cores have $m \in \{4, 6\}$: $m = 4$ for type B_2/C_2, present in B_n and F_4, and $m = 6$ for type G_2. Under this empirical classification, no equality occurs in simply-laced types, where rank-two parabolics have $m \le 3$.

This is an empirical observation, not a theorem or a fully scoped conjecture. We found the pattern in every exhaustively checked group (Table 1; witnesses include B_2 '1212', B_3 '2323', B_4 '3434', B_5 '4545', F_4 '2323', and G_2 '1212') and in the 64,944-interval seeded perturbation probe (Table 3). In that probe, every perturbation of an $m = 4$ dihedral core by extra cover steps that breaks the pure pattern strictly raises the ratio; the closest non-equality margins in the checked cases are 4–8%.

We have not attempted a proof. We also do not know whether the correct statement concerns the whole rank sequence $(1, 2, \dots, 2, 1)$, as in every recorded full witness, or only a local pattern around the equality index. The classification of short Bruhat intervals [13] may give a proved instance for intervals below a fixed length, but we have not carried out that argument.

Why Weyl groups escape the H_3 failure. The following is a possible mechanism, not a theorem. The H_3 counterexample in Example 2.2 contains an $m = 5$ dihedral core: the parabolic $\langle s_1, s_2\rangle \cong I_2(5)$ is part of the defining data of the failing interval. The crystallographic restriction excludes $m = 5$, and every $m \notin \{2, 3, 4, 6\}$, from rank-two parabolics in Weyl groups. Among rank-two parabolics with $m \ge 4$, only $m = 4$ and $m = 6$ remain. In Observation 7.1, these cores have ratio exactly 1, and none of the tested perturbations lowers the ratio below 1.

This pattern is consistent with the absence of an H_3-type failure in our exhaustive and near-top data. Because it relies on the empirical Observation 7.1, it does not prove that every Weyl-group interval is log-concave.

8. Discussion and open problems

We separate the conclusions by logical status.

Proved unconditionally: Theorem 4.1 verifies log-concavity for every Bruhat interval in A_2–A_7, B_2–B_6, D_4–D_6, E_6, F_4, and G_2, a total of 1,079,490,991 intervals. Theorem 4.3 and Proposition 4.7 establish exactly the stated near-top and sampled coverage; they do not verify the conjecture beyond those intervals. In type A, Theorem 6.1 proves the pointwise local limit bound and the stated fixed-window ratio law unconditionally for every $m \geq 4$, and Theorem 6.2 determines the global minimum exactly for every $4 \leq m \leq 560$. Theorem 5.5 proves the extremal inequality for rationally smooth lower intervals in irreducible simply-laced Weyl groups of rank ≤ 6, and in type A_{m-1} for $m \leq 17$ using the cited factorization result. Proposition 6.8 proves that (S1)–(S6) imply Conjecture 6.3.

Proved conditionally: Assuming Conjecture 6.3, Theorem 6.5 proves $\sigma^2(r_m - 1) = 1 - \frac{27}{25}m^{-1} + O(m^{-2})$ and the stated two-sided finite-m bound. Its upper-bound half is unconditional for every $m \geq 180$.

Conjectured or empirical: Brenti's conjecture remains open in general. Conjecture 5.1 is supported by exhaustive low-rank and near-top computations but is not proved for any infinite family of arbitrary intervals; Theorem 5.5 is its proved finite-range subclass. Conjecture 5.6 is also unproved in general. Observation 7.1 is an empirical equality classification, and the proposed explanation of the H_3 failure is not a proof.

Open: Conjecture 6.3 and the statements (S2)–(S6) remain open. Also open are Conjecture 5.1 in general rank, the rationally smooth cases beyond the stated ranges, a proved equality classification, the stronger central-location statement for the Mahonian minimum, and the E_8 cases.

For future work, these results provide reductions as well as data. If Conjecture 5.1 holds, verification of rank log-concavity in each irreducible, simply-laced Weyl group reduces from exponentially many intervals to the single Poincaré coefficient sequence of $[e, w_0]$; Theorem 5.5 establishes this comparison only for rationally smooth lower intervals in the stated finite ranges. In type A, that sequence is Mahonian, making classical local-limit methods available for the candidate extremal ratio. The equality data give a second structural target: the observed cores have $m \in \{4, 6\}$, both crystallographic, whereas Brenti's H_3 counterexample uses $m = 5$. This is a candidate explanation for the crystallographic boundary, not a proof. Finally, the exact-arithmetic logs make the finite computations reproducible, and Conjecture 6.3 is a named, explicitly quantified target for removing the condition from Theorem 6.5.

We next list several directions for further work.

(1) **Prove statements (S2)–(S6) of Conjecture 6.7, and hence Conjecture 6.3.** Proposition 6.8 gives this implication (together with the proved (S1)), and Theorem 6.5 would then make the sharp asymptotic unconditional. The far-region decay, crossover and mid-range bounds, variance floors and caps, reduction, and the whole range $4 \leq m \leq 560$ are already proved with explicit constants. The smallest computed margins occurred in the cumulant bounds of (S1) in the deepest band, where they are under 4%; that statement is now proved. Among the open statements, the missing ω-continuum coverage in (S5) and the uniform closure requirements of (S6) are where a proof effort should begin.

(2) **Extend Theorem 5.5 beyond rank 6.** Proving Conjecture 5.6 would also remove the $m \leq 17$ restriction in type A. The cases D_n for $n \geq 7$ and E_7, E_8 remain open even under the rationally-smooth restriction.

(3) **Self-dual and rank-symmetric Bruhat intervals** ([18], which classifies self-dual intervals) are a natural place to look for further explicit equality-case or extremal examples, complementing Observation 7.1.

(4) **Prove a finite-length case of Observation 7.1.** The classification of short Bruhat intervals [13] may yield an equality classification for intervals below some fixed length.

(5) E_8**.** We give no exhaustive or near-top data for E_8 ($|W| > 696$ billion). A thin near-top slab would extend the evidence for Conjecture 5.1 to every exceptional simply-laced type.

(6) **A quantitative form of Conjecture 2.11**, e.g. $\rho([u, v]) \geq 1 + c(W)$ with $c(W)$ explicit in simply-laced types decaying as predicted by Conjecture 5.1 and Theorem 6.1, would unify the extremal and asymptotic statements of this paper into a single sharp inequality.

Several related questions do not address Conjecture 2.11 itself. These include equivariant log-concavity of flag-variety cohomology [15] and equality cases for related inequalities [16, 17]. Chapelier-Laget and Fromentin [20] recently disproved a different conjecture from the same survey. The repository contains the query log for the literature search described in the Introduction.

Acknowledgments

Automated systems, principally Claude (Anthropic) in an agentic coding environment, wrote most of the verification code, drafted proof arguments for Sections 5 and 6, and helped assemble the manuscript from working notes. We subjected the draft proofs and numerical results to independent automated adversarial review passes (separate mathematics and numerics lanes), including a from-scratch numerical audit. These checks found and corrected at least three errors in automated drafts: a sign error, a false finite-range certificate, and a fabricated verification result.

Nikol Panayotova Savova and Sihao Huang selected the research questions and strategies, determined which statements were presented as theorems or conjectures, and independently checked the disclosed constants and counterexamples. They take responsibility for every claim and for any remaining error.

References

[1] F. Brenti, *Some open problems on Coxeter groups and unimodality*, arXiv:2410.09897.
[2] A. Björner and T. Ekedahl, *On the shape of Bruhat intervals*, Ann. of Math. **170** (2009), 799–817. arXiv:math/0508022.
[3] E. R. Canfield, S. Janson, and D. Zeilberger, *The Mahonian probability distribution on words is asymptotically normal*, Adv. Appl. Math. **46** (2011). arXiv:0908.2089.
[4] M. Bóna, *A combinatorial proof of the log-concavity of a famous sequence counting permutations*, Electron. J. Combin. **11(2)** (2004), #N2.
[5] S. G. Hoggar, *Chromatic polynomials and logarithmic concavity*, J. Combin. Theory Ser. B **16** (1974).
[6] W. Kook, *On the product of log-concave polynomials*, unpublished note, 2006.
[7] L. M. Butler, *The q-log-concavity of q-binomial coefficients*, J. Combin. Theory Ser. A **54** (1990), 54–63.
[8] B. E. Sagan, *Inductive proofs of q-log concavity*, Discrete Math. **99** (1992), 289–306.
[9] X.-T. Su, Y. Wang, and Y.-N. Yeh, *Unimodality problems of multinomial coefficients and symmetric functions*, Electron. J. Combin. **18(1)** (2011), #P73.
[10] G. Burrull, T. Gui, and H. Hu, *Asymptotic log-concavity of dominant lower Bruhat intervals via the Brunn–Minkowski inequality*, arXiv:2311.17980.
[11] J. B. Carrell, *The Bruhat order, algebraic cell decompositions and rationally smooth Schubert varieties*, Proc. Sympos. Pure Math. **56** (1994).
[12] V. Gasharov, *Factoring the Poincaré polynomials for the Bruhat order on S_n*, J. Combin. Theory Ser. A **83** (1998), 159–164.
[13] E. Akhmedova, *On Bruhat intervals of small lengths for Weyl groups*, arXiv:2110.00862.
[14] V. V. Petrov, *Sums of Independent Random Variables*, Springer, 1975.
[15] T. Gui, *On the equivariant log-concavity for the cohomology of the flag varieties*, arXiv:2205.05408.
[16] S. H. Chan and I. Pak, *Equality cases of the Stanley–Yan log-concave matroid inequality*, arXiv:2407.19608.
[17] R. van Handel, A. Yan, and X. Zeng, *The extremals of the Kahn–Saks inequality*, arXiv:2309.13434.
[18] C. Gaetz and Y. Gao, *Self-dual intervals in the Bruhat order*, Selecta Math. (2020). arXiv:2003.06710.
[19] A. Kessouri, M. Ahmia, H. Arslan, and S. Mesbahi, *Combinatorics of q-Mahonian numbers of type B and log-concavity*, arXiv:2408.02424.
[20] N. Chapelier-Laget and J. Fromentin, *A counterexample to a Brenti–Carnevale conjecture*, arXiv:2412.19593.